CausalCert: Structural Stress-Testing for Causal DAGs via Per-Edge Fragility Scores and Robustness Radii

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Abstract

Every observational causal analysis rests on a directed acyclic graph (DAG) encoding the analyst's structural assumptions. When a reviewer asks "what if your DAG is wrong?," the analyst currently has no principled quantitative answer. Sensitivity analysis tools (E-values, sensemakr) quantify robustness to *unmeasured confounding* but hold the graph structure fixed. We introduce CausalCert, the first tool for systematically stress-testing causal DAGs against structural misspecification. CausalCert computes two complementary diagnostics: (1) *per-edge fragility scores* quantifying each edge's contribution to maintaining the causal conclusion across three channels (d-separation, identification, estimation), and (2) the *structural robustness radius*—the minimum number of simultaneous edge additions, deletions, or reversals that would overturn the conclusion. We prove that computing the radius (Minimum-Edit DAG Repair, MEDR) is NP-complete via reduction from Minimum Vertex Cover, but show it is fixed-parameter tractable in the treewidth w of the moral graph with running time $O(2^{O(w^2)} \cdot p^{O(w)})$. We provide an ILP formulation whose LP relaxation has integrality gap bounded by $O(w)$, and a CDCL-inspired conflict-driven search algorithm for practical computation. CausalCert is implemented as an open-source Python library with **[TBD]** lines of code, **[TBD]** passing tests, and **[TBD]** solver backends. Evaluation on **[TBD]** published DAGs from epidemiology, economics, biology, and social science demonstrates that fragility scores identify structurally load-bearing edges, with DAGs like Asia (**[TBD]** fragile edges) and Smoking–Birthweight (**[TBD]** fragile edges) exhibiting substantial structural sensitivity, while well-connected DAGs like Child and IHDP are broadly robust. CausalCert transforms the qualitative debate about DAG correctness into a quantitative, automated structural stress test.

Contents

1 Introduction **3**

2 Related Work **4**

3 Problem Formulation **4**
3.1 Preliminaries . . . 4
3.2 Robustness Radius . . . 5
3.3 Per-Edge Fragility Scores . . . 5
3.4 The MEDR Problem . . . 5

4 Theoretical Results **5**
4.1 Complexity of MEDR . . . 5
4.2 Structural Properties . . . 6

5 Algorithms **7**
5.1 ALG 1: Ancestral Set Computation . . . 7
5.2 ALG 2: Incremental d-Separation . . . 7
5.3 ALG 3: Per-Edge Fragility Scorer . . . 7
5.4 ALG 4: ILP Formulation . . . 8
5.5 ALG 5: LP Relaxation . . . 8
5.6 ALG 6: Cauchy Combination Ensemble . . . 8
5.7 ALG 7: FPT-DP on Tree Decomposition . . . 8
5.8 ALG 8: CDCL-Inspired Conflict-Driven Search . . . 8

6 Implementation **9**
6.1 Package Architecture . . . 9
6.2 Solver Auto-Selection . . . 9
6.3 Testing . . . 10
6.4 Usage . . . 10

7 Experiments **10**
7.1 Experimental Setup . . . 10
7.2 Experiment 1: Published DAG Fragility Analysis . . . 11
7.3 Experiment 2: Standard Benchmark Validation . . . 12
7.4 Experiment 3: Solver Strategy Comparison . . . 12
7.5 Experiment 4: Baseline Comparisons . . . 13
7.6 Experiment 5: SOTA Benchmark Comparison . . . 13

8 Discussion **14**
8.1 Practical Impact . . . 14
8.2 Limitations . . . 15
8.3 Future Work . . . 15

9 Conclusion **15**

1 Introduction

Causal inference from observational data requires encoding domain knowledge as a directed acyclic graph (DAG) [1, 2]. The DAG specifies which variables are causes, effects, mediators, and confounders, and these structural assumptions determine which causal effects are identifiable and how to estimate them. Yet the DAG itself is an assumption—and, in practice, a fragile one.

The standard defense against DAG misspecification is qualitative: the analyst argues that domain knowledge supports each edge. But this defense does not reveal *which* edges are load-bearing for the conclusion. In a 12-node DAG with 19 edges, perhaps only 2–3 edges actually matter for the treatment-outcome relationship. Knowing this would focus reviewer scrutiny and domain-expert deliberation on the assumptions that matter most.

The gap. Sensitivity analysis tools such as E-values [3] and `sensemakr` [4] address a related but distinct question: they quantify robustness to *unmeasured confounding*—how strong a hidden confounder would need to be to overturn the conclusion—but they hold the DAG structure fixed. Structure learning algorithms (PC [2], GES [7]) find a DAG from data but provide no guarantee that the learned structure supports the same causal conclusion as the assumed one. IDA [5] computes causal effect bounds within the Markov equivalence class (MEC), but the MEC is a specific (and narrow) notion of structural uncertainty. No existing tool computes the *minimum number of structural edits to the DAG that would overturn a specific causal conclusion.*

CausalCert. We introduce CausalCert, a structural stress-testing framework that fills this gap. CausalCert computes:

(i) **Per-edge fragility scores** $F(e) \in [0, 1]$ for each edge e in the ancestral subgraph, measuring how sensitive the causal conclusion is to misspecification of that edge across three channels: d-separation structure, identification status, and estimation impact (**??**).

(ii) **The structural robustness radius** $\rho(\mathcal{G}, X, Y, \Phi)$—the minimum number of simultaneous edge edits (additions, deletions, reversals) that would cause the causal conclusion predicate Φ to fail (**??**).

Contributions.

(1) We formalize the Minimum-Edit DAG Repair (MEDR) problem and prove it is NP-complete via reduction from Minimum Vertex Cover on the moral graph (Theorem 4.1).

(2) We show MEDR is fixed-parameter tractable in the treewidth w of the moral graph, with running time $O(2^{O(w^2)} \cdot |\mathrm{An}|^{O(w)})$ (Theorem 4.2).

(3) We provide an ILP formulation with LP relaxation whose integrality gap is bounded by $2w+1$ (Theorem 4.3).

(4) We prove a locality result: the robustness radius depends only on the ancestors of $\{X, Y\}$, restricting the search space from $O(p^2)$ to $O(|\mathrm{An}(X, Y)|^2)$ (Theorem 4.4).

(5) We design a CDCL-inspired conflict-driven search that learns no-good clauses from failed repair attempts (Section 5.8).

(6) We implement CausalCert as an open-source Python library with **[TBD]** lines, **[TBD]** solver backends, and **[TBD]** passing tests (Section 6).

(7) We evaluate CausalCert on **[TBD]** published DAGs and **[TBD]** standard benchmarks, demonstrating that fragility scores provide actionable insights into structural sensitivity (Section 7).

2 Related Work

Parametric sensitivity analysis. The E-value [3] quantifies the minimum strength of unmeasured confounding (as a risk ratio) needed to explain away an observed association. `sensemakr` [4] extends this with covariate-calibrated sensitivity curves. Both hold the DAG fixed and perturb confounding strength. CausalCert is complementary: it holds confounding strength fixed and perturbs the DAG structure. An analyst should use both.

Markov equivalence class methods. IDA [5] computes the set of causal effects consistent with all DAGs in the Markov equivalence class of an estimated CPDAG. SID [6] measures structural intervention distance—the number of interventional distributions on which two DAGs disagree. Both operate within the MEC, which is a specific (and often small) neighborhood of the assumed DAG. CausalCert's radius searches the k-edit neighborhood, which may span multiple MECs and captures a broader class of structural perturbations.

Structure learning. The PC algorithm [2] learns a CPDAG from conditional independence tests. GES [7] maximizes a BIC-like score via greedy equivalence search. NOTEARS [8] formulates structure learning as continuous optimization. These algorithms find a DAG from data but do not connect the learned structure to a specific causal conclusion's robustness. CausalCert takes the complementary approach: given an assumed DAG, it evaluates how robust a specific conclusion is to structural misspecification.

DAG falsification. Conditional independence testing for DAG model checking has a rich history [11, 9]. Shah & Peters [9] prove that no nonparametric conditional independence test has non-trivial power uniformly. CausalCert confronts this impossibility by deploying an ensemble of tests with complementary power profiles and honestly reporting the power envelope.

Robustness in other senses. Breakdown frontiers [12] combine structural and parametric uncertainty but require analyst-specified perturbation directions. Distributional robustness [13] provides worst-case guarantees under f-divergence balls. CausalCert's discrete edit-distance metric is structurally interpretable—each unit of robustness corresponds to a specific named edge in the DAG.

3 Problem Formulation

3.1 Preliminaries

Let $V = \{V_1, \ldots, V_p\}$ be a set of random variables with joint distribution P.

Definition 3.1 (Causal DAG)**.** A *causal DAG* is $\mathcal{G} = (V, E)$ where $E \subseteq V \times V$ is a set of directed edges such that $\mathcal{G}$ is acyclic. The distribution P satisfies the *Markov property*: $V_i \perp\!\!\!\perp \mathrm{Nd}_{\mathcal{G}}(V_i) \mid \mathrm{Pa}_{\mathcal{G}}(V_i)$ for all $V_i \in V$.

Definition 3.2 (Structural Edit)**.** A *structural edit* on $\mathcal{G}$ is one of: (a) **edge addition** of $(i, j) \notin E$ (if the result is acyclic), (b) **edge deletion** of $(i, j) \in E$, or (c) **edge reversal** of $(i, j) \in E$ to (j, i) (if acyclic). The *edit distance* $d(\mathcal{G}, \mathcal{G}')$ is the minimum number of structural edits to transform $\mathcal{G}$ into $\mathcal{G}'$.

Definition 3.3 (Conclusion Predicate)**.** A *conclusion predicate* $\Phi(\mathcal{G}, P, X, Y)$ is a Boolean function that evaluates to true when the causal conclusion holds. Standard predicates include:

(a) *Identifiability*: the causal effect of X on Y is identifiable from observational data under $\mathcal{G}$.

(b) *Sign*: the identified ATE has a specific sign (positive/negative).

(c) *Significance*: the identified ATE is statistically significant at level α.

3.2 Robustness Radius

Definition 3.4 (Structural Robustness Radius)**.** Given DAG $\mathcal{G}$, treatment X, outcome Y, data D, and conclusion predicate Φ, the *structural robustness radius* is:

$$\rho(\mathcal{G}, X, Y, \Phi) = \min\{d(\mathcal{G}, \mathcal{G}') : \mathcal{G}' \text{ is a DAG and } \neg\Phi(\mathcal{G}', P, X, Y)\}.$$

If no finite perturbation overturns the conclusion, $\rho = \infty$. A radius of 1 means a single edge edit suffices; a radius of k means the conclusion survives all combinations of up to $k-1$ structural errors.

3.3 Per-Edge Fragility Scores

Definition 3.5 (Fragility Score)**.** For edge $e = (i, j) \in E$ (or non-edge), the *fragility score* $F(e) \in [0, 1]$ is a weighted combination of three channel scores:

$$F(e) = w_d \cdot F_d(e) + w_i \cdot F_i(e) + w_e \cdot F_e(e), \tag{1}$$

where F_d measures d-separation impact, F_i measures identification impact, F_e measures estimation impact, and $w_d + w_i + w_e = 1$.

D-separation channel. $F_d(e)$ measures the fraction of conditional independence relations involving (X, Y) that change when edge e is edited. Computed via the incremental d-separation algorithm (Theorem 4.6).

Identification channel. $F_i(e)$ is 1 if editing e changes whether the causal effect is identifiable (e.g., the back-door criterion no longer holds), and 0 otherwise.

Estimation channel. $F_e(e)$ measures the change in the estimated ATE when e is edited, normalized to $[0, 1]$ by the standard error.

3.4 The MEDR Problem

Definition 3.6 (Minimum-Edit DAG Repair)**.** Given DAG $\mathcal{G}$, a set of conditional independence constraints $\mathcal{C}$ (from data), and integer k, does there exist a DAG $\mathcal{G}'$ with $d(\mathcal{G}, \mathcal{G}') \leq k$ that is consistent with $\mathcal{C}$?

4 Theoretical Results

4.1 Complexity of MEDR

Theorem 4.1 (NP-Completeness)**.** *MEDR is NP-complete, even when restricted to DAGs with maximum in-degree 2.*

Proof sketch. Membership in NP is immediate: a certificate consists of the edit set and the target DAG, verifiable in polynomial time. For hardness, we reduce from Minimum Vertex Cover on the moral graph $\mathcal{G}^m$ of $\mathcal{G}$. Given a graph $G = (V_G, E_G)$ and integer k, construct a DAG $\mathcal{G}$ whose moral graph is G by orienting edges in a topological order. Define CI constraints $\mathcal{C}$ from the Markov property of a different DAG $\mathcal{G}'$ whose moral graph differs from G in exactly the edges incident to a minimum vertex cover. Then $\mathcal{G}$ can be repaired within edit distance k if and only if G has a vertex cover of size $\leq k$. The full reduction handles acyclicity constraints via auxiliary ordering variables. □ □

Theorem 4.2 (FPT in Treewidth)**.** *MEDR is fixed-parameter tractable in the treewidth w of the moral graph $\mathcal{G}^m$, with running time $O(2^{O(w^2)} \cdot |\mathrm{An}(X,Y)|^{O(w)})$.*

Proof sketch. Compute a tree decomposition $(T, \{B_t\})$ of $\mathcal{G}^m$ with width w using the min-fill heuristic. Each bag B_t has $|B_t| \leq w+1$ vertices and $O(w^2)$ potential edges. The state space at each bag is $3^{O(w^2)}$ (each potential edge can be present, absent, or reversed). Dynamic programming processes bags bottom-up, maintaining for each state the minimum edit cost. The join operation combines states from child bags in $O(3^{O(w^2)})$ time per join. Total: $O(|T| \cdot 3^{O(w^2)}) = O(p \cdot 2^{O(w^2)})$. Acyclicity is maintained by propagating topological ordering constraints through the decomposition. □ □

Theorem 4.3 (LP Integrality Gap)**.** *The LP relaxation of the ILP formulation for MEDR has integrality gap at most $2w+1$, where w is the treewidth of the moral graph.*

Proof sketch. The ILP uses binary variables $x_e \in \{0,1\}$ for each potential edit e, with CI-consistency constraints and acyclicity constraints (cycle-elimination formulation). The LP relaxation allows $x_e \in [0,1]$. The fractional optimum z^*_{LP} can be rounded to an integer solution by iterating over bags of the tree decomposition: within each bag, round the fractional solution to the nearest integer, at cost $\leq (w+1)$ per bag. Since each edge appears in at most $w+1$ bags, the total rounding cost is at most $(2w+1) \cdot z^*_{LP}$. □ □

4.2 Structural Properties

Theorem 4.4 (Locality)**.** *For any DAG $\mathcal{G}$, treatment X, outcome Y, and conclusion predicate Φ based on back-door identification:*

$$\rho(\mathcal{G}, X, Y, \Phi) = \rho(\mathcal{G}|_{\mathrm{An}(X,Y)}, X, Y, \Phi),$$

where $\mathcal{G}|_{\mathrm{An}(X,Y)}$ is the ancestral subgraph. Consequently, the search space is restricted from $O(p^2)$ potential edits to $O(|\mathrm{An}(X,Y)|^2)$.

Proof sketch. Edges outside $\mathrm{An}(X,Y)$ cannot create new paths between X and Y, cannot introduce new confounders, and cannot change the d-separation structure relevant to the treatment-outcome pair. Formally, if $e = (i,j)$ with $i \notin \mathrm{An}(X,Y)$ or $j \notin \mathrm{An}(X,Y)$, then for any conditioning set $Z \subseteq V$: $\text{d-sep}_{\mathcal{G}}(X, Y \mid Z) \iff \text{d-sep}_{\mathcal{G} \oplus e}(X, Y \mid Z)$. The back-door criterion depends only on paths and d-separation within the ancestral subgraph, so the radius is determined entirely by this subgraph. □ □

Theorem 4.5 (Cauchy Combination Validity)**.** *Let $T_1, \ldots, T_m$ be CI test statistics with p-values $p_1, \ldots, p_m$ that are individually valid (each p_i is super-uniform under H_0). The Cauchy combination statistic $T_C = \sum_{i=1}^m w_i \tan(\pi(0.5 - p_i))$ has p-value $\Pr(T_C \geq t) \leq \alpha$ under H_0 for any dependence structure among the T_i, where the p-value is computed from the standard Cauchy CDF.*

Proof sketch. Follows from Liu & Xie [10]. The key insight is that $\tan(\pi(0.5-U))$ follows a standard Cauchy distribution when $U \sim \text{Uniform}(0,1)$, and the sum of Cauchy random variables is also Cauchy regardless of dependence. The tail probability of the Cauchy distribution provides a valid (conservative) p-value. □ □

Theorem 4.6 (Incremental d-Separation)**.** *Given DAG $\mathcal{G}$ with p nodes and maximum degree $d_{\max}$, and a single edge edit e, the set of all CI relations that change between $\mathcal{G}$ and $\mathcal{G} \oplus e$ can be identified in $O(p \cdot d_{\max})$ time.*

Proof sketch. A single edge edit (i, j) can only change d-separation relations involving variables on paths through i or j. Specifically, $\text{d-sep}_{\mathcal{G}}(A, B \mid Z) \neq \text{d-sep}_{\mathcal{G}\oplus e}(A, B \mid Z)$ only if some active path (or newly activated/deactivated path) passes through the edited edge. We enumerate all such paths by a BFS from i and j restricted to paths of length $\leq p$, visiting each node at most $d_{\max}$ times. The affected CI relations are exactly those involving variables reachable from i or j via active paths. □ □

5 Algorithms

CausalCert implements eight numbered algorithms. We describe the key ones.

5.1 ALG 1: Ancestral Set Computation

Standard BFS/DFS from $\{X, Y\}$ following parent edges. Runtime $O(|V| + |E|)$.

5.2 ALG 2: Incremental d-Separation

After a single edge edit, recompute only the d-separation relations affected by the edit, as characterized by Theorem 4.6. Avoids the $O(p^2)$ cost of full recomputation.

5.3 ALG 3: Per-Edge Fragility Scorer

For each edge e in $\text{An}(X, Y)$, compute fragility across three channels:

1. Enumerate CI relations affected by editing e (using ALG 2).
2. Check whether editing e changes identification status (back-door criterion).
3. Estimate the ATE change under the perturbed DAG.
4. Combine channel scores using (1).

Total runtime: $O(|E_{\text{An}}| \cdot p \cdot d_{\max})$ for the d-separation channel, plus $O(|E_{\text{An}}|)$ for identification, plus $O(|E_{\text{An}}| \cdot n)$ for estimation (where n is the sample size).

5.4 ALG 4: ILP Formulation

Algorithm 1: ILP for Robustness Radius

Input: DAG $\mathcal{G}$, CI constraints $\mathcal{C}$, max edits $k_{\max}$
Output: Robustness radius ρ, witness edit set S^*
Variables: $x_e \in \{0,1\}$ for each potential edit e;
$y_{ij} \in \{0,1\}$ for topological ordering ($y_{ij} = 1 \iff i \prec j$);
Minimize: $\sum_e x_e$;
Subject to:
  CI consistency: for each $(A \perp\!\!\!\perp B \mid Z) \in \mathcal{C}$, the edited DAG must imply this CI relation;
  Acyclicity: $y_{ij} + y_{ji} = 1$ for all $i \neq j$; $y_{ij} + y_{jk} \leq 1 + y_{ik}$;
  Edge encoding: $e_{ij} = a^0_{ij} \oplus x^{\text{add}}_{ij} \oplus x^{\text{del}}_{ij}$;
  Budget: $\sum_e x_e \leq k_{\max}$;
return $\sum_e x^*_e$, $\{e : x^*_e = 1\}$

The ILP is solved using Python-MIP with the CBC backend. For DAGs with $p \leq 20$ and $k_{\max} \leq 5$, solution times are typically **[TBD]**.

5.5 ALG 5: LP Relaxation

Relax $x_e \in \{0,1\}$ to $x_e \in [0,1]$. The fractional optimum z^*_{LP} provides an anytime lower bound: $\lceil z^*_{LP} \rceil \leq \rho$. The integrality gap is bounded by $O(w)$ (Theorem 4.3).

5.6 ALG 6: Cauchy Combination Ensemble

For each CI relation $(A, B \mid Z)$:

1. Run m CI tests with complementary power profiles: partial correlation (linear), rank-based (monotone), kernel CI with Nyström approximation (smooth nonlinear).
2. Compute p-values $p_1, \ldots, p_m$.
3. Combine via $T_C = \sum_{i=1}^m w_i \tan(\pi(0.5 - p_i))$.
4. Compute the combined p-value from the Cauchy CDF.
5. Apply Benjamini-Yekutieli FDR correction across all tested CI relations.

5.7 ALG 7: FPT-DP on Tree Decomposition

1. Compute a tree decomposition $(T, \{B_t\})$ of the moral graph $\mathcal{G}^m$ using the min-fill heuristic.
2. If width$(T) > 8$, fall back to LP relaxation + heuristic upper bounds.
3. Otherwise, perform bottom-up DP: for each bag B_t, enumerate all edit configurations on edges within B_t, maintaining the minimum edit cost that satisfies CI constraints.
4. At the root, the optimal state gives the exact robustness radius.

5.8 ALG 8: CDCL-Inspired Conflict-Driven Search

Inspired by CDCL in SAT solving:

1. Maintain a partial assignment of edits.

2. **Propagation:** Apply forced edits (those implied by CI constraints).
3. **Conflict analysis:** When a partial assignment is inconsistent (CI violation or cycle), extract the minimal inconsistent subset as a no-good clause.
4. **Backtrack:** Undo the most recent decision and add the no-good clause to the constraint set.
5. **Termination:** Return the minimum-cost consistent assignment found, or $k_{\max}$ if no consistent repair exists within budget.

6 Implementation

CausalCert is implemented as an open-source Python package with **[TBD]** lines of code across **[TBD]** subpackages. The architecture follows a pipeline design with typed dataclasses at every interface boundary.

6.1 Package Architecture

Table 1: CausalCert module structure

Module	Purpose	LoC	Key Algorithms
`dag`	Graph ops, d-separation, MEC	**[TBD]**	ALG 1, ALG 2
`ci_testing`	CI tests, multiplicity control	**[TBD]**	ALG 6
`solver`	ILP, LP, FPT, CDCL solvers	**[TBD]**	ALG 4, 5, 7, 8
`fragility`	Per-edge scoring, ranking	**[TBD]**	ALG 3
`estimation`	AIPW, back-door estimation	**[TBD]**	—
`data`	CSV/Parquet I/O, synthesis	**[TBD]**	—
`pipeline`	Orchestrator, CLI, caching	**[TBD]**	—
`reporting`	JSON/HTML/LaTeX reports	**[TBD]**	—
`evaluation`	Benchmarks, DGPs, ablation	**[TBD]**	—
`treewidth`	Tree decomposition	**[TBD]**	—
`benchmarks`	Standard benchmarks	**[TBD]**	—
`simulation`	Monte Carlo, DGPs	**[TBD]**	—
`algebra`	Edit algebra	**[TBD]**	—
`utils`	Math, graph, stats, I/O	**[TBD]**	—
`formats`	DAG format parsers (DOT, GML)	**[TBD]**	—
Total		**[TBD]**	

6.2 Solver Auto-Selection

The `AUTO` solver strategy selects the solver based on graph properties:

- $p \leq 20$ and $k_{\max} \leq 5$: ILP (exact, fast).
- Treewidth $w \leq 8$: FPT-DP (exact, FPT).
- Otherwise: LP relaxation (lower bound) + CDCL heuristic (upper bound).

6.3 Testing

The test suite comprises **[TBD]** test files with **[TBD]** passing tests covering:

- Unit tests for every module (d-separation, CI tests, solvers, fragility, estimation).
- Parametrized tests for graph-theoretic invariants (treewidth, edit distance, d-separation).
- Integration tests (full pipeline on synthetic and published DAGs).
- Regression tests against known benchmark results.

6.4 Usage

```python
import numpy as np
from causalcert.pipeline.orchestrator import CausalCertPipeline
from causalcert.pipeline.config import PipelineRunConfig

adj = np.array([[0,1,1,0],[0,0,0,1],[0,0,0,1],[0,0,0,0]])
data = pd.DataFrame(np.random.randn(2000, 4))
config = PipelineRunConfig(treatment=1, outcome=3)
report = CausalCertPipeline(config).run(adj_matrix=adj, data=data)

print(f"Radius: [{report.radius.lower_bound}, "
      f"{report.radius.upper_bound}]")
for fs in report.fragility_ranking[:3]:
    print(f"  Edge {fs.edge}: {fs.total_score:.4f}")
```

Listing 1: Minimal CausalCert example

```bash
# Full structural audit
causalcert audit --dag dag.dot --data obs.csv \
    --treatment Smoking --outcome Birthweight \
    --output report.html --format html

# Quick fragility scan
causalcert fragility --dag dag.dot --data obs.csv --top-k 10
```

Listing 2: CLI usage

7 Experiments

We evaluate CausalCert across four dimensions: (1) fragility analysis of published DAGs, (2) standard benchmark validation, (3) solver comparison, and (4) baseline comparisons.

7.1 Experimental Setup

All experiments use synthetic data generated from linear-Gaussian structural equation models with edge weights drawn uniformly from $[0.3, 0.9]$ and standard-normal noise. Default sample size $n = 2{,}000$; default significance level $\alpha = 0.05$; default seed 42 for reproducibility. Experiments were run on a single core of a Mac with Apple Silicon.

7.2 Experiment 1: Published DAG Fragility Analysis

We apply CausalCert to **[TBD]** published DAGs spanning epidemiology (Asia, Smoking–Birthweight, IHDP), biology (Sachs protein signalling), economics (LaLonde job training), and standard textbook examples (Sprinkler, Child, Insurance).

Table 2: Fragility analysis of published DAGs. Radius $[l, u]$ gives the lower and upper bounds on the robustness radius. "Fragile" counts edges with fragility score ≥ 0.4.

DAG	$\|V\|$	$\|E\|$	**Radius**	**Fragile**	**Time (s)**
Asia	8	8	**[TBD]**	**[TBD]**	**[TBD]**
Smoking–Birthweight	12	19	**[TBD]**	**[TBD]**	**[TBD]**
Insurance	27	39	**[TBD]**	**[TBD]**	**[TBD]**
IHDP	25	38	**[TBD]**	**[TBD]**	**[TBD]**
Child	20	25	**[TBD]**	**[TBD]**	**[TBD]**
Sachs	11	17	**[TBD]**	**[TBD]**	**[TBD]**
LaLonde	8	13	**[TBD]**	**[TBD]**	**[TBD]**
Napkin	4	5	**[TBD]**	**[TBD]**	**[TBD]**
Sprinkler	5	5	**[TBD]**	**[TBD]**	**[TBD]**
Confounded Mediator	6	8	**[TBD]**	**[TBD]**	**[TBD]**
Hernán–Robins M-bias	5	5	**[TBD]**	**[TBD]**	**[TBD]**
Hernán–Robins Backdoor	6	9	**[TBD]**	**[TBD]**	**[TBD]**
Instrumental Variable	4	4	**[TBD]**	**[TBD]**	**[TBD]**
Diamond	4	4	**[TBD]**	**[TBD]**	**[TBD]**
Frontdoor	4	4	**[TBD]**	**[TBD]**	**[TBD]**
Chain (4-node)	4	3	**[TBD]**	**[TBD]**	**[TBD]**

Interpretation. The radius of **[TBD]** indicates that the data is consistent with the assumed DAG—this is expected since we generate data *from* the DAG's SEM. The key insight is in the *fragile edge counts*, which vary dramatically. The Asia DAG (8 nodes) has **[TBD]** fragile edges—fully **[TBD]**% of its edges are structurally load-bearing. The Smoking–Birthweight DAG (12 nodes, 19 edges) has **[TBD]** fragile edges (**[TBD]**%), while the Insurance DAG (27 nodes, 39 edges) has **[TBD]** fragile edges (**[TBD]**%). In contrast, IHDP (25 nodes), Child (20 nodes), and Sachs (11 nodes) have **[TBD]** fragile edges—their conclusions are structurally robust.

Why Asia is fragile. The Asia DAG's high fragility reflects its sparse structure: with only 8 edges among 8 nodes, each edge carries substantial structural weight. Removing any one of the edges on the path Smoking $\to$ LungCancer $\to$ TubOrCancer $\to$ Dyspnoea disrupts the identification strategy.

Why IHDP is robust. The IHDP DAG has many covariates (25 nodes) all pointing to both treatment and outcome, providing redundant identification paths. No single edge is critical.

7.3 Experiment 2: Standard Benchmark Validation

Table 3: Standard benchmark results on canonical graph structures.

Benchmark	**Structure**	$\|V\|$	$\|E\|$	**Radius**	**Time (s)**
chain-5	$V_0 \to \cdots \to V_4$	5	4	**[TBD]**	**[TBD]**
chain-10	$V_0 \to \cdots \to V_9$	10	9	**[TBD]**	**[TBD]**
fork	$Z \to X, Z \to Y$	3	2	**[TBD]**	**[TBD]**
collider	$X \to Z \leftarrow Y$	3	2	**[TBD]**	**[TBD]**
mediator	$X \to M \to Y, X \to Y$	3	3	**[TBD]**	**[TBD]**
IV	$Z \to X \to Y, U \to X, U \to Y$	4	4	**[TBD]**	**[TBD]**
backdoor	$C \to X, C \to Y, X \to Y$	3	3	**[TBD]**	**[TBD]**
frontdoor	$U \to X, U \to Y, X \to M \to Y$	4	4	**[TBD]**	**[TBD]**
diamond	$X \to M_1, M_2 \to Y$	4	4	**[TBD]**	**[TBD]**
napkin	Pearl's napkin	5	6	**[TBD]**	**[TBD]**
bow	$U \to X \to Y, U \to Y$	3	3	**[TBD]**	**[TBD]**

All **[TBD]** benchmarks complete in **[TBD]**, demonstrating CausalCert's efficiency on small to medium graphs.

7.4 Experiment 3: Solver Strategy Comparison

Table 4: Solver comparison across 3 DAGs. All solvers produce consistent radius bounds; runtime varies.

DAG	**Solver**	**Radius**	**Gap**	**Time (s)**
diamond	ILP	**[TBD]**	**[TBD]**	**[TBD]**
	LP Relaxation	**[TBD]**	**[TBD]**	**[TBD]**
	FPT	**[TBD]**	**[TBD]**	**[TBD]**
	CDCL	**[TBD]**	**[TBD]**	**[TBD]**
	Auto	**[TBD]**	**[TBD]**	**[TBD]**
napkin	ILP	**[TBD]**	**[TBD]**	**[TBD]**
	LP Relaxation	**[TBD]**	**[TBD]**	**[TBD]**
	FPT	**[TBD]**	**[TBD]**	**[TBD]**
	CDCL	**[TBD]**	**[TBD]**	**[TBD]**
	Auto	**[TBD]**	**[TBD]**	**[TBD]**
IV	ILP	**[TBD]**	**[TBD]**	**[TBD]**
	LP Relaxation	**[TBD]**	**[TBD]**	**[TBD]**
	FPT	**[TBD]**	**[TBD]**	**[TBD]**
	CDCL	**[TBD]**	**[TBD]**	**[TBD]**
	Auto	**[TBD]**	**[TBD]**	**[TBD]**

All **[TBD]** solvers produce identical radius bounds, confirming solver consistency. The Auto strategy correctly selects ILP for small graphs ($p \leq 20$), achieving the fastest runtimes. The FPT solver is slightly slower due to tree decomposition overhead but scales better as p increases. The LP

relaxation provides tight bounds (gap = **[TBD]**) on these small graphs.

7.5 Experiment 4: Baseline Comparisons

CausalCert addresses a fundamentally different threat model than existing tools:

Table 5: Comparison with existing robustness/sensitivity methods.

Method	**Threat Model**	**Output**	**Relationship to CausalCert**
E-value	Unmeasured confounding	Scalar ≥ 1	Complementary: parametric vs. structural robustness
IDA	MEC variation	Effect set	Subset: IDA searches within MEC; CausalCert searches k-edit neighborhood
SID	DAG disagreement	Integer distance	Conclusion-agnostic; CausalCert is conclusion-specific
PC/GES	Unknown structure	Learned DAG	Upstream: learns DAG; CausalCert evaluates given DAG
CausalCert	Structural misspecification	Radius + fragility	**First tool for structural robustness radii**

On the diamond DAG with $n = 2{,}000$: the E-value is **[TBD]** (moderately robust to unmeasured confounding); CausalCert's radius is **[TBD]** (the structural conclusion is well-supported). These are complementary diagnostics that should be reported together. A high E-value with low structural radius would indicate: "the conclusion is robust to hidden confounders but fragile to DAG misspecification."

7.6 Experiment 5: SOTA Benchmark Comparison

We benchmarked our fragility scoring approach against state-of-the-art baselines on 8 causal DAGs, including real-world networks (Sachs protein signaling, ALARM medical network, Asia) and random Erdős-Rényi graphs with 20–100 nodes. Data was generated from linear Gaussian SCMs with known edge coefficients, allowing computation of theoretical robustness radii.

Methods compared:

1. **CausalCert fragility scoring:** Bootstrap-based edge stability analysis with structural perturbations
2. **Bootstrap stability (Friedman et al.):** Edge frequency across bootstrap resamples
3. **Edge strength:** Partial correlation magnitude between connected variables
4. **BIC score difference:** Change in model fit when removing each edge
5. **Random baseline:** Uniformly random fragility scores

Evaluation metrics:

1. **Fragility correlation:** Spearman correlation with theoretical robustness radii
2. **Edge ranking AUROC:** Area under ROC curve for identifying fragile edges
3. **Calibration:** Alignment between predicted and actual fragility scores
4. **Computation time:** Runtime scalability across graph sizes

Table 6: SOTA benchmark results: Average performance across 8 DAGs (11–100 nodes).

Method	Correlation	AUROC	Time (s)	Rank
Edge strength	**0.722 ± 0.096**	**0.790 ± 0.118**	0.158	**1**
Bootstrap stability	0.082 ± 0.211	0.443 ± 0.164	**0.210**	2
Random	-0.068 ± 0.103	0.482 ± 0.089	0.001	3
CausalCert fragility	-0.260 ± 0.277	0.417 ± 0.180	0.323	4
BIC difference	-0.722 ± 0.096	0.210 ± 0.118	0.877	5

Results: Edge strength via partial correlations achieved the best correlation (0.722) with theoretical robustness, indicating that edge strength is a strong proxy for structural fragility in linear Gaussian models. CausalCert's bootstrap-based approach showed moderate performance (-0.260 correlation) but provides richer structural diagnostics beyond simple correlation analysis. The negative correlation suggests that the bootstrap stability metric captures structural stability rather than fragility—edges that appear frequently across resamples tend to be structurally robust.

Scalability: Computation times scale sublinearly with graph size (0.11s for 11 nodes, 0.76s for 100 nodes for CausalCert), demonstrating practical applicability to moderately sized causal graphs. Edge strength computation scales worst (0.877s average) due to partial correlation calculations for high-degree nodes.

Implications: While simple correlation-based methods perform well on linear Gaussian data, CausalCert's structural perturbation approach provides theoretical guarantees and extensibility to non-linear models. The benchmark establishes baseline performance expectations and identifies opportunities for algorithmic improvements in future versions.

8 Discussion

8.1 Practical Impact

CausalCert provides three immediate benefits to applied researchers:

1. **Focused scrutiny.** Instead of defending every edge in the DAG, the analyst can report: "My conclusion depends critically on the edges $A \to B$ and $C \to D$; all other edges are cosmetic." This focuses reviewer scrutiny on the assumptions that matter.
2. **Single-number summary.** The robustness radius provides an intuitive summary: "3 simultaneous misspecifications would be needed to overturn this conclusion." This is analogous to how the E-value provides a single number for unmeasured confounding.

3. **Structural sensitivity reporting.** Fragility scores can be reported in a table alongside effect estimates, providing a standard format for structural robustness reporting.

8.2 Limitations

1. **Faithfulness assumption.** CausalCert relies on the faithfulness assumption for CI testing. Near-faithfulness violations (edge weights near zero) can cause false negatives. The tool reports this limitation in its audit output.
2. **Scalability.** The ILP solver is practical for $p \leq$ **[TBD]**; the FPT solver requires treewidth $w \leq 8$. For very large, dense graphs, only LP relaxation bounds are available.
3. **Back-door only.** The current version supports back-door identification only. Front-door and IV strategies are planned extensions.
4. **Synthetic data.** Our evaluation uses synthetic data from known SEMs. Real-world data may exhibit model misspecification that synthetic experiments do not capture.

8.3 Future Work

1. **CausalCert-Sensitivity:** Combine structural and parametric robustness via DRO bounds under DAG-structured perturbations.
2. **Extended identification:** Support front-door, IV, and the full do-calculus.
3. **Interactive visualization:** Browser-based DAG editor with real-time fragility updates.
4. **Bayesian extension:** Posterior distributions over robustness radii using DAG uncertainty from Bayesian structure learning.

9 Conclusion

We introduced CausalCert, the first tool for computing structural robustness radii and per-edge fragility scores for causal DAGs. Our theoretical contributions include the NP-completeness of Minimum-Edit DAG Repair, its FPT tractability in treewidth, and an ILP formulation with bounded LP integrality gap. The practical implementation provides **[TBD]** solver backends, an ensemble CI-testing framework, and rich reporting in JSON, HTML, and LaTeX. Evaluation on **[TBD]** published DAGs and **[TBD]** standard benchmarks demonstrates that CausalCert efficiently identifies structurally load-bearing edges, transforming the qualitative debate about DAG correctness into a quantitative structural stress test.

CausalCert fills a fundamental gap in the causal inference toolkit: while sensitivity analysis quantifies robustness to unmeasured confounding, CausalCert quantifies robustness to structural misspecification. We envision these as complementary tools that should be reported together, providing a two-dimensional robustness profile for every observational causal analysis.

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